F1 Fluid Mechanics I: Classical Flows

Level: from Navier-Stokes to graduate

One of ten documents. Reading order and syllabus: `CHAPTER_ORGANIZATION.md`; plan and status: `ROADMAP.md`.

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1 Continuum hypothesis and kinematics of flow

TODO

TODO(F1/ch01): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(CM1/ch02); TODO(M1/ch04)

2 Conservation laws: mass, momentum, energy

TODO

TODO(F1/ch02): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch01); M1:16 (The Gauss (divergence) theorem)

3 The Navier-Stokes equations and constitutive relations

TODO

TODO(F1/ch03): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch02)

4 Ideal flow: Euler equations and Bernoulli

TODO

TODO(F1/ch04): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch03)

5 TODO(F1/ch05)

Prerequisites

Part M1:II; Part M1:III; M1:13 (Integration of 1-forms); M2:11 (Torsion, the Levi-Civita connection and geodesics)

5.1 Fluids: circulation needs a metric

The velocity field u is a vector (it generates the flow). Its circulation $\Gamma = \oint_\gamma u^\flat$ integrates the covector $u^\flat = g(u, \cdot)$, so it needs a metric. The vorticity is the 2-form $\mathrm{d}u^\flat$ (the "curl vector" $\operatorname{curl} u = (\star \mathrm{d}u^\flat)^\sharp$ again needs g and the orientation). For a barotropic ideal fluid with enthalpy h,

Euler's equation reads

$$\partial_t u^\flat + \iota_u \mathrm{d}u^\flat = -\mathrm{d}\big(h + \tfrac{1}{2}|u|^2\big). \tag{5.1}$$

Where this equation comes from (steps). (1) Euler: $\partial_t u + (u\cdot\nabla)u = -\nabla h$. (2) Vector identity: $(u\cdot\nabla)u = \nabla(\tfrac{1}{2}|u|^2) - u\times\omega$ with $\omega = \operatorname{curl} u$. (3) Lower the index: $(u\times\omega)^\flat = -\iota_u \mathrm{d}u^\flat$ (same triple-product computation as for the Lorentz force, with $\mathrm{d}u^\flat = \iota_\omega \mathrm{vol}$). Substituting gives the stated equation.

Kelvin's theorem (steps). By Cartan's formula (M2:7.2), $\mathcal{L}_u u^\flat = \iota_u \mathrm{d}u^\flat + \mathrm{d}(\iota_u u^\flat) = \iota_u \mathrm{d}u^\flat + \mathrm{d}(|u|^2)$. Adding $\mathrm{d}(|u|^2)$ to both sides of Euler's equation:

$$(\partial_t + \mathcal{L}_u)u^\flat = \mathrm{d}\big(\tfrac{1}{2}|u|^2 - h\big),$$

an exact form. For a loop $\gamma(t)$ carried by the flow, $\frac{\mathrm{d}}{\mathrm{d}t}\oint_{\gamma(t)} u^\flat = \oint_{\gamma(t)} (\partial_t + \mathcal{L}_u)u^\flat = \oint \mathrm{d}(\cdots) = 0$, because exact forms integrate to zero on closed loops (Section M1:13.5). Kelvin's circulation theorem is Stokes plus this observation (Arnold–Khesin [13]).

6 Potential flow and complex potentials

TODO

TODO(F1/ch06): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch04); M2:17 (The residue theorem as Stokes' theorem)

7 Viscous flows: exact solutions, Stokes flow, lubrication

TODO

TODO(F1/ch07): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch03)

8 Boundary layers

TODO

TODO(F1/ch08): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch07)

9 Hydrodynamic stability

TODO

TODO(F1/ch09): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch08)

10 Introduction to turbulence

TODO

TODO(F1/ch10): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch09)

11 Compressible flow and gas dynamics

TODO

TODO(F1/ch11): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch04)

12 Waves in fluids

TODO

TODO(F1/ch12): this chapter has no content yet. Its place in the reading order is listed in `CHAPTER_ORGANIZATION.md`.

Prerequisites: TODO(F1/ch06)

References

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